

Customer Analytics Dashboard using Machine Learning

Project Report

Abstract

This project focuses on developing an interactive Customer Analytics Dashboard using Python, Machine Learning, and Streamlit to analyze customer behavior and generate business insights. The system applies RFM analysis, K-Means clustering, Customer Lifetime Value analysis, churn risk identification, and AI-powered recommendations to help businesses improve customer engagement and revenue optimization.

Introduction

Customer analytics helps organizations understand customer purchasing behavior and improve marketing strategies. This project was developed to provide an intelligent analytics platform capable of customer segmentation and business insight generation through interactive visualizations and machine learning techniques.

Objectives

- Perform customer segmentation using RFM analysis and K-Means clustering.
- Analyze customer purchasing behavior and spending patterns.
- Build an interactive dashboard using Streamlit.
- Generate AI-powered business insights using Groq API.
- Deploy the project online using Streamlit Cloud.

Technologies Used

- Python
- Pandas and NumPy
- Scikit-learn
- Streamlit
- Plotly
- Matplotlib and Seaborn
- Groq API
- GitHub and Streamlit Cloud

Dataset Description

The project uses the Online Retail dataset containing customer transaction records such as invoice details, quantities, prices, customer IDs, and countries. Additional features including Recency, Frequency, Monetary value, Customer Lifetime Value, and customer personas were generated during preprocessing.

Methodology

- Data cleaning and preprocessing
- RFM feature engineering
- Data normalization using StandardScaler
- Customer segmentation using K-Means clustering
- Dashboard development using Streamlit
- AI-powered recommendation generation

RFM Analysis

The RFM methodology serves as a strategic framework utilized to assess and quantify individual consumer engagement patterns.

Recency (R)

This metric quantifies the duration elapsed since the most recent transactional activity performed by a client.

Diminished recency values signify a base of highly engaged and active consumers.

Frequency (F)

This attribute evaluates the regularity and number of purchase events recorded for each unique identifier.

Elevated frequency levels are indicative of established brand loyalty and consistent retention.

Monetary (M)

This dimension calculates the cumulative financial expenditure generated by a customer during the analyzed period.

Greater monetary contributions highlight the most financially significant and premium segments.

Implementation of the RFM framework facilitates the identification of:

- Steadfast and devoted clientele
- Primary revenue-generating accounts
- Dormant or non-participatory users
- Segments susceptible to imminent attrition

Customer Segmentation

The K-Means clustering technique is implemented as an unsupervised machine learning methodology to categorize consumer profiles into homogenous groupings.

Operational Procedures

- Extraction of relevant RFM metrics
- Data standardization utilizing the StandardScaler algorithm
- Execution of the K-Means clustering model
- Allocation of specific customer personas based on behavioral clusters

Categorical Personas

Cluster ID	Persona Definition
0	VIP Customers
1	Loyal Customers
2	At Risk Customers
3	New Customers

Elbow Methodology

The elbow heuristic was applied to identify the most appropriate number of distinct clusters for analysis.

Silhouette Index Evaluation

Silhouette coefficients were computed to assess the statistical robustness and separation of the resulting clusters.

Customer Lifetime Value (CLV) Analysis

Customer Lifetime Value serves as a predictive metric to estimate the aggregate financial contribution generated by a specific client throughout their tenure.

Computational framework applied:

$CLV = \text{Product of Frequency and Monetary metrics}$

Utilization of the CLV framework enables organizations to:

- Distinguish high-worth consumer segments
- Refine and enhance retention methodologies
- Rationalize and optimize promotional expenditures

Churn Risk Analysis

Consumer profiles characterized by elevated recency metrics are categorized as segments susceptible to imminent attrition.

The analytical interface distinguishes high-attrition risk accounts and formulates strategic re-engagement methodologies.

Illustrative intervention strategies include:

- Targeted promotional discount campaigns
- Customized and personalized incentives
- Client retention loyalty reward programs

Data Preprocessing

Ensuring analytical accuracy necessitates a rigorous data preprocessing phase.

The following refinement procedures were implemented:

Handling Missing Values

Observations containing null data points were eliminated from the dataset.

Removing Invalid Transactions

Records exhibiting the following attributes:

- Quantities less than or equal to zero
- Unit prices less than or equal to zero

were systematically purged to maintain data integrity.

Date Conversion

The original InvoiceDate attribute was transformed into a standardized datetime object.

Total Price Calculation

A derived TotalPrice metric was engineered:

TotalPrice = Quantity multiplied by UnitPrice

Dashboard Features

- Dark-themed interactive dashboard
- KPI cards for customer analytics
- Interactive filters for persona, country, and revenue
- 3D customer segmentation visualization
- Correlation heatmap
- Top customer analysis
- Downloadable reports
- AI-generated business insights

AI-Powered Insights

The project integrates Groq API to generate automated business insights and recommendations. The AI module analyzes customer metrics and provides suggestions related to customer retention, marketing strategies, churn reduction, and revenue optimization.

Results and Observations

- VIP customers contribute the highest revenue.
- Loyal customers show high purchase frequency.
- At-risk customers require re-engagement strategies.
- Interactive filtering improves analytical flexibility.
- AI insights enhance business decision-making.

Advantages of the System

- Interactive and user-friendly interface
- Real-time customer analysis
- Improved customer segmentation
- Business intelligence support
- Cloud-based deployment accessibility

Future Enhancements

- Predictive churn modeling
- Recommendation systems
- Database integration
- Natural language query support
- Mobile-responsive interface
- Automated PDF report generation

Conclusion

The Customer Analytics Dashboard successfully demonstrates the integration of Machine Learning, Data Visualization, and AI-powered analytics for customer behavior analysis. The system provides valuable business insights and supports data-driven decision-making through an interactive and scalable analytics platform.